Chapter 3. Continuous Functions

Limits and continuity, extreme and intermediate values, uniform continuity, monotone functions

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Throughout, $X \subset \mathbb{R}^n$, $Y \subset \mathbb{R}^m$, and functions $f : X \to Y$.

3.1 Limits and continuity

Motivation. The ϵ–δ and sequential definitions of continuity agree, so Chapters 1–2 apply to functions.

Let x_0 be a limit point of X. We write $\lim_{x\to x_0} f(x) = y_0$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - y_0\| < \epsilon$ whenever $x \in X$ and $0 < \|x - x_0\| < \delta$. f is **continuous at** $x_0 \in X$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(x_0)\| < \epsilon$ whenever $x \in X$, $\|x - x_0\| < \delta$; equivalently, x_0 is isolated in X, or $\lim_{x\to x_0} f(x) = f(x_0)$. f is **continuous** if it is continuous at every point.

Proposition 3.1 (3.1.1)

If $\lim_{x\to x_0} f(x) = y_0$ and $\lim_{x\to x_0} g(x) = z_0$, then $f + g \to y_0 + z_0$, $af \to ay_0$, and for real-valued functions $fg \to y_0 z_0$, and $f/g \to y_0/z_0$ if $z_0 \neq 0$.

Proof. As in Theorem 1.14, or by Proposition 3.3 below applied to sequences. ■

Proposition 3.2 (3.1.2)

If $f, g : X \to \mathbb{R}^m$ are continuous, so is $af + bg$; for real-valued f, g, so are fg and f/g (where $g \neq 0$).

Proof. Apply Proposition 3.1 at each non-isolated point. ■

Polynomials in several variables are therefore continuous, and so are rational functions where the denominator does not vanish.

Proposition 3.3 (3.1.3, sequential criterion)

f is continuous at x_0 iff $f(x_k) \to f(x_0)$ for every sequence $x_k \to x_0$ in X.

Proof. ($\Rightarrow$) Given ϵ, take δ; for large k, $\|x_k - x_0\| < \delta$, so $\|f(x_k) - f(x_0)\| < \epsilon$.

($\Leftarrow$) If f is not continuous at x_0, for some ϵ and each k there is x_k with $\|x_k - x_0\| < 1/k$ but $\|f(x_k) - f(x_0)\| \geq \epsilon$. ■

Similarly $\lim_{x\to x_0} f(x) = y_0$ iff $f(x_k) \to y_0$ for every sequence $x_k \to x_0$ with $x_k \neq x_0$.

Proposition 3.4 (3.1.4)

If $f : X \to Y$ is continuous at x_0 and $g : Y \to Z$ is continuous at $f(x_0)$, then $g \circ f$ is continuous at x_0.

Proof. Given ϵ, choose η for g at $f(x_0)$ and then δ for f at x_0 with η in place of ϵ. ■

Example 3.1

(a) $f(x) = \sin(1/x)$ $(x \neq 0)$ has no limit at 0: $x_k = 1/(k\pi + \pi/2)$ gives ± 1 alternately.
(b) $x \sin(1/x) \to 0$ by the squeeze $|x \sin(1/x)| \leq |x|$.
(c) Thomae's function ($f(p/q) = 1/q$ in lowest terms, $f = 0$ at irrationals) is continuous exactly at the irrationals.

(d) $f(x,y) = xy/(x^2+y^2)$, $f(0,0) = 0$, is continuous along every line through the origin, but $f(t,t) = \frac{1}{2}$, so f is not continuous at 0.

Theorem 3.5 (3.1.5)

For $f : X \to Y$ the following are equivalent:
(a) f is continuous;
(b) $f^{-1}(V)$ is open in X for every open V in Y;
(c) $f^{-1}(E)$ is closed in X for every closed E in Y.

Proof. (a)⇒(b): if $x \in f^{-1}(V)$ and $N(f(x),\epsilon) \cap Y \subset V$, the δ of continuity gives $N(x,\delta) \cap X \subset f^{-1}(V)$. (b)⇒(a): $f^{-1}(N(f(x_0),\epsilon) \cap Y)$ is open in X and contains x_0. (b)⇔(c): $f^{-1}(Y - E) = X - f^{-1}(E)$. ■

3.2 Continuity, compactness and connectedness

Motivation. Extreme and intermediate values express that continuity preserves compactness and connectedness, in any $\mathbb{R}^n$.

Theorem 3.6 (3.2.1)

If $f : X \to Y$ is continuous and surjective and X is compact, then Y is compact.

Proof. Let $\{U_i\}$ be an open cover of Y; by Theorem 3.5 and Proposition 2.6, $f^{-1}(U_i) = V_i \cap X$ with V_i open in $\mathbb{R}^n$, and the V_i cover X. Finitely many $V_{i_1}, \ldots, V_{i_k}$ do; then $U_{i_1}, \ldots, U_{i_k}$ cover $Y = f(X)$. ■

Corollary 3.7 (3.2.2, extreme value theorem)

A continuous $f : X \to \mathbb{R}$ on a compact set attains its maximum and minimum.

Proof. $f(X)$ is compact, hence bounded and closed; so $\sup f(X)$ is finite and, being a limit of points of $f(X)$, lies in $f(X)$. ■

Corollary 3.8 (3.2.3)

If X is compact and $f : X \to \mathbb{R}$ is continuous with $f > 0$, then there is $\delta > 0$ with $f(x) \geq \delta$ for all x.

Proof. Take $\delta = \min f > 0$. ■

Corollary 3.9 (3.2.4)

A continuous $f : [a,b] \to \mathbb{R}$ attains its maximum and minimum.

Corollary 3.10 (3.2.5)

If X is compact and $f : X \to Y$ is a continuous bijection, then f^{-1} is continuous.

Proof. Let U be open in X. $X - U$ is closed in the compact X, so compact (Proposition 2.24); then $f(X - U) = Y - f(U)$ is compact, hence closed, so $f(U)$ is open in Y. By Theorem 3.5, f^{-1} is continuous. ■

Compactness is needed: $t \mapsto (\cos t, \sin t)$ is a continuous bijection $[0, 2\pi) \to S^1$ whose inverse is not continuous. For nonempty $A, B \subset \mathbb{R}^n$ let $\operatorname{dist}(A, B) = \inf\{\|a - b\| : a \in A, b \in B\}$.

Proposition 3.11 (3.2.6)

If A is compact, B is closed and $A \cap B = \emptyset$, then $\operatorname{dist}(A, B) > 0$.

Proof. The function $d(x) = \operatorname{dist}(\{x\}, B)$ is continuous (indeed $|d(x) - d(y)| \le \|x - y\|$) and positive on A, since $\mathbb{R}^n - B$ is open. By Corollary 3.8 it is bounded below on A by some $\delta > 0$. ■

The compactness is needed: the closed sets $\{(x, 0)\}$ and $\{(x, 1/x) : x > 0\}$ are at distance 0.

Lemma 3.12 (3.2.7)

S is disconnected iff there is a continuous surjection $f : S \to \{0, 1\}$.

Proof. If U, V disconnect S, let $f = 0$ on $U \cap S$ and 1 on $V \cap S$; preimages of subsets of $\{0, 1\}$ are relatively open, so f is continuous. Conversely, $f^{-1}(0)$ and $f^{-1}(1)$ are nonempty and relatively open (Theorem 3.5), so they are $U \cap S$ and $V \cap S$ with U, V open (Proposition 2.6); then $U \cap V \cap S = f^{-1}(0) \cap f^{-1}(1) = \emptyset$ and $S \subset U \cup V$. ■

Theorem 3.13 (3.2.8)

If $f : X \to Y$ is continuous and surjective and X is connected, then Y is connected.

Proof. If Y were disconnected, Lemma 3.12 gives a continuous surjection $g : Y \to \{0, 1\}$, and then $g \circ f : X \to \{0, 1\}$ is a continuous surjection. ■

Corollary 3.14 (3.2.9)

If $I \subset \mathbb{R}$ is an interval and $f : I \to \mathbb{R}$ is continuous, then $f(I)$ is an interval.

Proof. Proposition 2.28 and Theorem 3.13. ■

Corollary 3.15 (3.2.10, intermediate value theorem)

If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < \alpha < f(b)$, then $f(c) = \alpha$ for some $c \in (a, b)$.

Corollary 3.16 (3.2.11)

If $f : [a, b] \to \mathbb{R}$ is continuous, then $f([a, b])$ is a bounded closed interval $[m, M]$.

Proof. It is an interval (Corollary 3.14) and compact (Theorem 3.6), containing its minimum m and maximum M. ■

A function mapping intervals to intervals need not be continuous, for example $\sin(1/x)$ extended by 0 at 0.

Corollary 3.17 (3.2.12, fixed point)

Every continuous $f : [a, b] \to [a, b]$ has a fixed point.

Proof. $g(x) = x - f(x)$ has $g(a) \le 0 \le g(b)$; apply Corollary 3.15 (or $g(a) = 0$ or $g(b) = 0$). ■

Example 3.2

Every polynomial of odd degree has a real root (it takes both signs). Every $a > 0$ has a unique positive n-th root, since x^n is continuous, strictly increasing and unbounded on $[0, \infty)$.

3.3 Uniform continuity

Motivation. One δ for all points is what integration needs; on compact sets continuity gives it.

$f : X \to Y$ is **uniformly continuous** if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(y)\| < \epsilon$ whenever $x, y \in X$, $\|x - y\| < \delta$.

Theorem 3.18 (3.3.1, Heine)

A continuous function on a compact set is uniformly continuous.

Proof. Given ϵ, choose for each $x \in X$ a $\delta(x) > 0$ with $\|f(y)-f(x)\| < \epsilon/2$ for $y \in X$, $\|y-x\| < \delta(x)$. The sets $N(x, \delta(x)/2)$ cover X; finitely many with centres $x_1, \dots, x_k$ do. Let $\delta = \min \delta(x_i)/2$. If $\|x - y\| < \delta$, then $\|x - x_i\| < \delta(x_i)/2$ for some i, and $\|y - x_i\| < \delta(x_i)$; so $\|f(x) - f(y)\| \le \|f(x) - f(x_i)\| + \|f(x_i) - f(y)\| < \epsilon$. ■

Example 3.3

x^2 is not uniformly continuous on $\mathbb{R}$ ($(n+1/n)^2 - n^2 > 2$), and $1/x$ is not uniformly continuous on $(0, 1)$; $\sqrt{x}$ is uniformly continuous on $[0, \infty)$ since $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$.

Theorem 3.19 (3.3.2)

A uniformly continuous function maps Cauchy sequences to Cauchy sequences.

Proof. Given ϵ, take δ from uniform continuity and N with $\|x_m - x_n\| < \delta$ for $m, n \ge N$. ■

Corollary 3.20 (3.3.3)

A uniformly continuous $f : X \to \mathbb{R}^m$ extends uniquely to a continuous function on $\overline{X}$, which is again uniformly continuous.

Proof. For $x_0 \in \overline{X}$ and $X \ni x_k \to x_0$, $(f(x_k))$ is Cauchy (Theorem 3.19), hence convergent (Theorem 2.14); let $F(x_0)$ be the limit, independent of the sequence (interlace two), with $F = f$ on X. If $\|x - y\| < \delta$ in $\overline{X}$, take $x_k \to x$, $y_k \to y$ with $\|x_k - y_k\| < \delta$: $\|F(x) - F(y)\| = \lim \|f(x_k) - f(y_k)\| \le \epsilon$. Uniqueness follows from Proposition 3.3. ■

3.4 Monotone functions

Motivation. Monotone functions have only jump discontinuities, and countably many.

For $f:(a,b)\to\mathbb{R}$ write $f(x+)=\lim_{t\to x^+}f(t)$ and $f(x-)=\lim_{t\to x^-}f(t)$ when these exist.

Theorem 3.21 (3.4.1)

A monotone $f:(a,b)\to\mathbb{R}$ has one-sided limits at every point, and $f(a+)$, $f(b-)$ exist in $[-\infty,\infty]$. If f is increasing and $a<x<y<b$, then $f(x-)\le f(x)\le f(x+)\le f(y-)\le f(y)\le f(y+)$, and $f(x+)=\inf_{t>x}f(t)$, $f(x-)=\sup_{t<x}f(t)$.

Proof. Let f increase. $A=\{f(t):a<t<x\}$ is bounded above by $f(x)$; let $s=\sup A$. Given ϵ, some $f(t_0)>s-\epsilon$, and for $t_0<t<x$, $s-\epsilon<f(t_0)\le f(t)\le s$; so $f(x-)=s\le f(x)$. Similarly $f(x+)=\inf_{t>x}f(t)\ge f(x)$. For $x<y$, choosing $x<t<y$, $f(x+)\le f(t)\le f(y-)$. ■

Corollary 3.22 (3.4.2)

The set of discontinuities of a monotone function on an interval is countable.

Proof. For increasing f, f is discontinuous at x iff $f(x-)<f(x+)$. Choose a rational $r(x)\in(f(x-),f(x+))$; by Theorem 3.21 these open intervals are disjoint for distinct x, so $x\mapsto r(x)$ is injective into $\mathbb{Q}$. ■

Conversely, for any countable set $\{x_k\}\subset(a,b)$, $f(x)=\sum_{x_k<x}2^{-k}$ is increasing and discontinuous exactly at the x_k.

Theorem 3.23 (3.4.3)

For a continuous $f:I\to\mathbb{R}$ on an interval, f is injective iff it is strictly monotone. In that case $f(I)$ is an interval and $f^{-1}:f(I)\to I$ is continuous.

Proof. Strictly monotone implies injective. Conversely, if f is injective but not strictly monotone, some $x<y<z$ in I have, say, $f(y)>\max(f(x),f(z))$; any $\alpha\in(\max(f(x),f(z)),f(y))$ is attained in (x,y) and in (y,z) (Corollary 3.15), a contradiction. For strictly increasing f, f^{-1} is strictly increasing on the interval $f(I)$ (Corollary 3.14) with image the interval I; a jump $g(y-)<g(y+)$ would omit values (Theorem 3.21), so f^{-1} is continuous. ■

3.5 Selected exercises

Motivation. Four book exercises using density, compactness, connectedness and completeness.

$F\subset\mathbb{R}^n$ is closed iff it contains the limits of its convergent sequences (Proposition 2.3(b), Theorem 2.4).

Exercise 3.5.2 (Cauchy's functional equation)

Let $f:\mathbb{R}\to\mathbb{R}$ be continuous with $f(x+y)=f(x)+f(y)$ for all x,y. Show that $f(x)=ax$ for all x, where $a=f(1)$.

Solution.

Step 1. $f(0)=f(0+0)=2f(0)$, so $f(0)=0$; then $0=f(r+(-r))=f(r)+f(-r)$ gives $f(-r)=-f(r)$.

Step 2. $f((n+1)r) = f(nr) + f(r)$ gives $f(nr) = nf(r)$ for $n \in \mathbb{N}$ by induction, and for $n \in \mathbb{Z}$ by Step 1.

Step 3. $a = f(n \cdot \frac{1}{n}) = nf(\frac{1}{n})$, so $f(\frac{m}{n}) = mf(\frac{1}{n}) = a\frac{m}{n}$: $f(q) = aq$ on $\mathbb{Q}$.

Step 4. For $x \in \mathbb{R}$ take $q_k \in \mathbb{Q} \cap (x, x+1/k)$ (Theorem 1.8); by Proposition 3.3, $f(x) = \lim aq_k = ax$. ■

Exercise 3.5.4 (Closed graphs)

For $f : \mathbb{R} \to \mathbb{R}$ let $G = \{(x, f(x)) : x \in \mathbb{R}\} \subset \mathbb{R}^2$ be its graph.
(a) Show that if f is continuous, then G is closed.
(b) Show that if f is bounded and G is closed, then f is continuous.
(c) Show by an example that boundedness cannot be dropped in (b).

Solution.

(a) If $(x_k, f(x_k)) \to (x, y)$, then $x_k \to x$, so $y = \lim f(x_k) = f(x)$ (Proposition 3.3) and $(x, y) \in G$.

(b)

Step 1. Let $|f| \le M$ and f be discontinuous at x: there are $\epsilon > 0$, $x_k \to x$ with $|f(x_k) - f(x)| \ge \epsilon$ (Proposition 3.3).

Step 2. A subsequence $f(x_{k_j}) \to y$ (Theorem 2.10), with $|y - f(x)| \ge \epsilon$.

Step 3. $(x_{k_j}, f(x_{k_j})) \to (x, y) \in G$, so $y = f(x)$, contradicting Step 2.

(c) $f(x) = 1/x$ $(x \neq 0)$, $f(0) = 0$ is discontinuous at 0 with closed graph: let $(x_k, f(x_k)) \to (x, y)$.

Step 1. $x \neq 0$: $f(x_k) = 1/x_k \to 1/x$ for large k, so $y = f(x)$.

Step 2. $x = 0$: $|f(x_k)| \le K$, so $x_k \neq 0$ forces $|x_k| \ge 1/K$; as $x_k \to 0$, $x_k = 0$ eventually and $y = 0 = f(0)$. ■

Exercise 3.5.9 (Antipodal points with equal values)

Let $T = \{(\cos t, \sin t) : t \in \mathbb{R}\} \subset \mathbb{R}^2$ be the unit circle and $f : T \to \mathbb{R}$ continuous. Show that $f(x) = f(-x)$ for some $x \in T$.

Solution.

Step 1. T is the continuous image of $[0, 2\pi]$, hence connected (Proposition 2.28, Theorem 3.13).

Step 2. $g(x) = f(x) - f(-x)$ is continuous on T (Propositions 3.2, 3.4), so $g(T)$ is an interval (Theorem 3.13, Proposition 2.28).

Step 3. For $p = (1, 0)$, $g(-p) = -g(p)$, so $0 \in g(T)$: some x has $f(x) = f(-x)$. ■

Exercise 3.5.10 (Contraction mapping principle)

Let $X \subset \mathbb{R}^n$ be nonempty and closed, $0 < r < 1$, and $f : X \to X$ with $\|f(x) - f(y)\| \le r\|x - y\|$ for all $x, y \in X$.
(a) Show that f is uniformly continuous.
(b) For $x_0 \in X$ let $x_{k+1} = f(x_k)$. Show that (x_k) is a Cauchy sequence.
(c) Show that f has exactly one fixed point x, that $x_k \to x$, and that $\|x_k - x\| \le \frac{r^k}{1-r}\|x_1 - x_0\|$.

Solution.

(a) $\delta = \epsilon$: $\|x - y\| < \delta \Rightarrow \|f(x) - f(y)\| \le r\|x - y\| < \epsilon$.

(b)

Step 1. By induction, $\|x_{j+1} - x_j\| = \|f(x_j) - f(x_{j-1})\| \le r\|x_j - x_{j-1}\| \le \cdots \le r^j\|x_1 - x_0\|$.

Step 2. For $m > k$, by the triangle inequality,

$$\|x_m - x_k\| \le \sum_{j=k}^{m-1} \|x_{j+1} - x_j\| \le \|x_1 - x_0\| \sum_{j=k}^{\infty} r^j = \frac{r^k}{1-r}\|x_1 - x_0\|.$$

Step 3. $r^k \to 0$ (Example 1.1(b)), so (x_k) is Cauchy.

(c)

Step 1. $x_k \to x \in \mathbb{R}^n$ (Theorem 2.14), and $x \in X$ as X is closed.

Step 2. By (a), $f(x) = \lim f(x_k) = \lim x_{k+1} = x$ (Proposition 3.3).

Step 3. $f(y) = y$ gives $\|x - y\| \le r\|x - y\|$, so $y = x$.

Step 4. $m \to \infty$ in Step 2 of (b) gives $\|x - x_k\| \le \frac{r^k}{1-r}\|x_1 - x_0\|$.

Closedness is needed: $f(x) = x/2$ on $X = (0, 1]$ is a contraction without a fixed point. ■